• Summarization:

1-Descriptive statistics : it's about summarizing and presenting data in a meaningful way

2-Inferential statistics : it's about making inferences and predictions about population based on sample of data

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•Descriptive:

1-Measures of central tendency (mean - mode – median)

•Mean : basically it's the average of an array = (sum /n)
n is number of values

•Mode : is the most repeated number in values

•Median : in a sorted array ,it's the middle value that divides the array into 2 equal size arrays

-case 1 : (array size is odd)
Median = n//2 (floor division)

-case 2 : (array size is even)
Median = average of 2 middle values
=((n/2)-1 + (n/2))/2

2-Measures of variability (Range - variance - standard deviation)

•**Range** : it's value depends on the maximum and minimum values of the array that's why it's not really efficient to measure the variation

-Range = max -min

•**Deviation** : the deviation of an entry x in an array is the difference between the entry and mean of the array

-Deviation = $\sum(X - M)^2$

•**Variance : σ^2** population variance of a data set

-unlike range variance's value depends on the value of all data entries (deviation)

- $\sigma^2 = \sum(X - M)^2 / n$

-but, the variance's unit is the square of data's unit therefore it's not the best measure of variation neither

•Standard deviation (STD) **σ** : the square root of variance , so this measure solve the units problem of variance

-Population STD = $\frac{\sqrt{\{\sum(\mathbf{X} - \mathbf{M})^2\}}}{n}$

-Sample STD = $\frac{\sqrt{\{\sum(\mathbf{x}-\bar{\mathbf{x}})^2\}}}{\boldsymbol{n}-\mathbf{1}}$, $\bar{\mathbf{x}}$ is mean of a sample of data from the population

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3- Quartiles :

-a measure that divides data into 4 parts each part has 25% of the data

- used to analyze distribution and variability of a data set {Q1 ,Q2, Q3} the divisors

• How to calculate {Q1 ,Q2, Q3}

First we divide data into 2 parts ,then we take each half and divide it into 2 quarters so we get 4 quarters

- Q2 : the middle value that divides array into 2 = Median

- Q1 = Median for first half

- Q3 = Median for second half

• **Interquartile Range (IQR) =** (Q3 - Q1)

-this range focuses on the middle where the extreme values are

- **Example : calculate (mean-mode-median- σ^2- std – quartiles - IQR)**

[5 5 7 , 16 29 30] , n=6

Mean=92/6 =15.34

Mode= most repeated=5

Median: n is even , sorted array

Median=(7+16)/2=11.5

Range=30-5=25

σ^2=

$$\frac{\{(5-15.34)^2+(5-15.34)^2+(7-15.34)^2+(16-15.34)^2+(29-15.34)^2+(30-15.34)^2)}{6}$$

=80.84

STD = $\sqrt{\{\sigma^2\}}$ = 8.9

Q2= median =11.5

Q1 =mid of 1st half =5

Q3 =mid of 2nd half =29

IQR =Q3 -Q1=24